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Note: Please refer to the Python notebook for executable code, extensive commentary, and results for each question in this document.

Step 1 - Defining the purpose of training, validation, and testing datasets

1a - Purpose of the Training Dataset

The training dataset covers the stretch from October 2000 to November 2014 — about 14 years of market history. At the training step, our models learn the underlying patterns, relationships, and market behaviors that will shape their understanding. In many ways, this period forms the foundation of our entire modeling process. The characteristics of these years directly influence how stable and reliable the model will be, especially when applied to risk management. That is why it is important to look closely at the key macroeconomic trends, structural shifts, and market forces that dominated during this time.

The training dataset serves three purposes in financial engineering:

1. **Establish Baseline Relationships:** Identify how macroeconomic indicators, market variables, and geopolitical factors interact with crude oil prices over time.
2. **Train the Model to identify Regimes:** To improve its ability to predict future conditions, let the model experience a variety of market environments, including bullish, bearish, and volatile phases.
3. **Build a Parameter Framework:** Using historical data, fit the parameters of statistical or machine learning models, such as conditional dependencies in Bayesian networks or transition probabilities in Hidden Markov models.

By selecting a training window that includes both good and bad(downturn) periods, we ensure that the model's parameter estimates are influenced by a range of market conditions rather than a single dominant trend. The training dataset is the most important input for our model calibration step in this project. The model will forecast or categorize market regimes and assess risk measures in Steps 4–7. These subsequent stages run the risk of becoming unreliable in the absence of a strong training set since the model may pick up patterns that are either too general (underfitting) or too specific (overfitting). For example,

a. The training set for a Hidden Markov Model enables the estimation of state transition probabilities, such as the possibility of switching from a "low-volatility" regime to a "high-volatility" regime.

b. It gives the empirical foundation for calculating conditional probabilities in a Bayesian Network context, such as the likelihood that the price of oil will fall given concurrent increases in VIX and DXY.
c. In a VaR or Expected Shortfall context, it determines the statistical distribution parameters that drive future loss estimates.

Why this period matters

The selected period captures several major economic cycles, crises, and structural shifts in the world monetary markets. The dataset includes the post-dot-com-bubble slowdown in 2002, the commodity boom from 2003 to 2007, and the GFC (Global Financial Crisis) of 2008, with its recovery phase in 2009. The chosen period also captures the monetary policy changing frameworks of 2010-2011 and the early indication of the commodity slump in 2014-2016. All of these events impact the crude oil prices, volatility indices, exchange rates, and macroeconomic indicators. From a modeling perspective, having a mix of strong years and tough ones is actually a plus. By being exposed to both calm and turbulent market conditions, the training dataset gives the model a broader view of reality. Those periods of extreme shocks — combined with the longer training window — help the model spot the deeper, more reliable links between macroeconomic trends and oil price movements.

Key market features in the training set

The training set captures a range of market features that together paint a vivid picture of the period.

1. It includes WTI crude oil prices and returns, which reflect both the steep plunge during the global financial crisis and the strong upward run from 2003 to 2008.
2. The Volatility Index (VIX) serves as a barometer of market fear — spiking during crises, often in step with sharp declines in oil demand.
3. The U.S. Dollar Index (DXY) also plays a role: whenever the dollar surged, as it did during the 2008 flight to safety, oil prices generally came under pressure.
4. On the broader economic front, measures like GDP growth, CPI inflation, and refinery utilization rates offer insight into the underlying forces shaping oil demand.
5. The Energy Uncertainty Index (EUI) and Geopolitical Risk Index (GPR) serve as markers of market tension, spiking during turbulent periods such as the Arab Spring or the 2003 Iraq War — times when geopolitical unrest cast a long shadow over global energy markets.

These variables provide complementary signals from the standpoint of modeling. These kinds of multi-variable interactions are essential for training regime-switching HMM or Bayesian Network models.

1b - Purpose of the Validation Dataset

The validation dataset runs from December 2014 through April 2019, covering roughly four and a half years. We think of this stage as a kind of "checkpoint" in the modeling process. The training set is where we shape the model's parameters, but the validation set's job is different — here, we tweak and adjust, checking how well the model's insights hold up when faced with data it has not seen before. In finance, this step is especially important because it is a balancing act: we want the model to be complex enough to capture useful patterns, yet simple enough to avoid the trap of overfitting. That is when a model can look brilliant on paper, but struggle badly in the real world with fresh data. From a practical perspective, the validation dataset has three major roles that closely reflect how financial researchers and practitioners work with models in the real world.

1. **Hyperparameter Optimization:** For methods like Hidden Markov Models, the validation set lets us experiment — tweaking the number of regimes or adjusting how we kick off the computations. If we are using Bayesian Networks, this step helps us compare scoring rules like BIC or K2, so we can spot the model structure that fits best without simply echoing the quirks of the training sample.

2. **Out-of-Sample Risk Calibration:** Think of the validation set as a kind of "stress test." We use it to check how calculations of Value-at-Risk (VaR) and Expected Shortfall (ES) perform when the model confronts market conditions it has not seen before. Risk calibration offers a crucial reality check: if the model starts flagging problems early here, it is a warning that future shocks might throw off our risk forecasts.

3. **Sensitivity and Stability Testing:** Validation is not just about numbers matching up —it is also about change. By comparing macroeconomic or geopolitical relationships during this period to those in training, we can spot when patterns shift meaningfully, which is common in commodities and fast-changing markets. This flags structural breaks and helps us decide if our model's foundations are still solid, or if it needs to adjust to new realities.

Why this period matters

The validation period offers a fascinating cross-section of oil market dynamics and the broader macroeconomic backdrop. In late 2014, crude prices went into a freefall — a consequence of surging U.S. shale output colliding with OPEC's surprise decision to keep production steady. That shock rippled through the market, eventually bottoming out in early 2016 when commodity prices hit multi-year lows. Then came a partial rebound between 2017 and 2018, supported by coordinated OPEC supply cuts — only to run into another headwind as U.S.–China trade tensions began to dominate sentiment.

For us, this sequence is more than just historical context; it is a stress test for the model. The challenge is clear: can the model detect and correctly classify these regime shifts, moving seamlessly from low-volatility to high-volatility environments, without losing predictive accuracy? This period, with its alternating cycles of panic and recovery, is exactly the kind of real-world complexity that determines whether a model is robust — or just fitted to the past.

Key market features in the validation set

Looking at the validation period, a few market features stand out as particularly relevant for stress-testing the model.

1. WTI prices and returns showcase the post-2014 recovery, punctuated by brief bullish runs that followed extended downturns — exactly the kind of shifts that test whether the model can detect regime transitions after prolonged weakness.
2. VIX levels were mostly subdued compared to the 2008 crisis. Still, they did flare up during corrections in 2015 and again in late 2018, providing a good check on the model's ability to capture volatility clustering.
3. DXY movements added another layer of complexity: the dollar's strength in 2015–2016 weighed on oil prices, reminding us that currency effects can be just as influential as supply-demand shifts in commodity pricing.
4. On the macro and geopolitical front, GDP growth moved within a modest range. However, the Geopolitical Risk Index (GPR) and Energy Uncertainty Index (EUI) still captured moments of heightened tension — from U.S.–Iran disputes to the broader market jitters after the 2016 Brexit vote.

Overall, this dataset reflects a transitional phase in the oil market — shifting from the structural oversupply shock of 2014 toward a more policy-driven rebalancing, which is what makes it such a rich testing set. The mix of volatility regimes forces model tuning to happen under real-world market complexity, building confidence that the final framework will hold up in the testing phase, where robustness matters most.

1c - Purpose of the Testing Dataset

The testing dataset spans July 2019 to September 2023, giving us just over four years of fresh market data — data the model has never **seen** during either training or validation. This is where theory meets reality. Up until now, our model has learned from the training dataset and been fine-tuned with the validation dataset. The testing dataset is the final proving ground — the place where we find out if all that preparation really translates into predictive power and robustness in the real world. In financial engineering terms, this is where we measure true out-of-sample performance under genuinely unfamiliar market regimes. For us, the testing dataset serves three main purposes:

1. **Final Performance Assessment:** We test whether the model can hold its ground in new market conditions — maintaining accuracy, stability, and interpretability. This means checking the regime classification precision for our HMMs and the conditional probability accuracy for Bayesian Networks.
2. **True Out-of-Sample Risk Estimation:** We evaluate whether Value-at-Risk (VaR) and Expected Shortfall (ES) estimates stay realistic under fresh shocks — without the safety net of last-minute parameter tuning.
3. **Robustness Under Extreme Events:** Since this window overlaps with one of the most disruptive macroeconomic events in recent history — COVID-19 — we assess the model's ability to handle high-impact, low-probability events and still function effectively.

Why This Period Matters

The testing window captures some of the most chaotic and unpredictable conditions the oil market has faced in decades. The COVID-19 pandemic brought an unprecedented collapse in oil demand, driving WTI prices briefly into negative territory in April 2020 — a once-unthinkable anomaly in energy markets. The rebound was swift but uneven: 2021 saw prices surge during global reopening, only for them to spike again in early 2022 amid the Russia–Ukraine conflict. This geopolitical shock triggered a broader energy crisis, pushing commodity markets into a sustained high-volatility phase.

In the aftermath, 2022–2023 marked a moderation period shaped by tightening monetary policy, stubborn inflation, and shifting OPEC+ production strategies — a combination that fundamentally altered market dynamics. From a modeling perspective, this period is priceless because it forces us to deal with:

a. **Non-linear shocks** — from the pandemic-driven collapse to the rapid rebound.
b. **Persistent high volatility** — as seen during the 2022 energy crisis.
c. **Macro-policy shifts** — like aggressive Fed and ECB rate hikes.
d. **Structural supply-demand changes** — influenced by accelerated energy transition policies.

Key Market Features in the Testing Set

WTI Prices & Returns: From the historic dip into negative prices in 2020 to extreme upward swings, this dataset tests the model's ability to cope with tail-risk events.

1. **VIX:** Sharp volatility spikes in March–April 2020 and early 2022 challenge the model's capacity to capture volatility clustering.
2. **DXY:** Dollar strength in 2022–2023 puts the model's crude price forecasts under added currency-driven pressure.
3. **GPR & EUI:** Elevated geopolitical and energy risk scores during the Russia–Ukraine war allow us to evaluate how well these signals remain predictive in high-risk energy markets.

4. **Macro Indicators (GDP, CPI):** High inflation and uneven economic recovery patterns test adaptability to shifting macro baselines.

In summary, the testing dataset is essentially our final testing conditions — a combination of structural policy shifts, extended periods of high volatility, and one of market shocks. It is exactly the type of environment where weaker models tend to break down. If our framework can navigate these conditions successfully, we can be confident that it is more than just an academic construct. It becomes a practical, market-ready tool, which can deliver insights that stand up in real-world decision-making.

To conclude, our dataset spans October 2000 to September 2023 and is split into three key phases — training, validation, and testing — each with a distinct role in the modeling process. The training set (2000–2014) forms the foundation, exposing the model to a wide range of market environments: the post-dot-com slowdown, the 2003–2007 commodity boom, the 2008 financial crisis, and the run-up to the 2014 oil price collapse. This diversity allows the model to learn how crude oil prices interact with macroeconomic indicators and geopolitical events in both stable and stressed conditions.

The validation set (2014–2019) serves as our checkpoint phase, where we fine-tune the model without overfitting — whether that is selecting the right number of regimes in HMMs or finding the optimal Bayesian Network structure. This period covers the sharp 2014–2016 oil price crash, the partial OPEC-driven recovery, and shifts triggered by U.S.–China trade tensions, making it an ideal stress test for regime detection and out-of-sample risk calibration.

Finally, the testing set (2019–2023) is the ultimate testing ground. It places the model in front of market events it has never “seen” before — from the COVID-19 demand collapse and April 2020’s negative WTI prices to the 2021–2022 rebound and volatility spikes from the Russia–Ukraine conflict. These extreme, policy-driven, and structurally shifting conditions allow us to assess if the framework can hold up outside the lab. Structuring the dataset this way mirrors best practice in financial engineering — ensuring a clean separation between learning, tuning, and evaluation, and testing the model where it matters most: in real-world decision-making.

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Step 2 - Comparing validation and testing datasets

To follow best practices in financial engineering, we split the dataset into three non-overlapping segments — training, validation, and testing — each with a clear role in model development and evaluation.

Dataset Name	Start Date	End Date	Duration	Purpose
Training	October 31, 2000	November 30, 2014	~14 years	Learn relationships and parameters from diverse market regimes.
Validation	December 31 2014	April 30, 2019	~4.5 years	Optimize hyperparameters and evaluate structural stability.
Testing	July 31, 2019	September 30, 2023	~4.2 years	Assessment of real-world robustness under an unseen scenario.

The training set spans a variety of macroeconomic cycles and oil market regimes, giving the model exposure to both stable and volatile periods. The validation set acts as a mid-course reality check, helping us fine-tune parameters without overfitting. The testing set presents new, often extreme events, providing a true out-of-sample test of both predictive accuracy and risk assessment. This structure ensures every stage draws from an independent, chronologically consistent dataset — mirroring the way models are built, tuned, and then tested on future, unseen market conditions before real-world deployment.

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Step 3 - Validating the model

Purpose and Approach in Alvi (2018)

In Section 4.3 of his dissertation, Alvi (2018) focuses on validating probabilistic graphical models — namely Hidden Markov Models (HMMs) and Bayesian Networks (BNs) — for forecasting crude oil price regimes. His goal was to test whether these models could reliably detect shifts between high- and low-volatility states, not only in the historical training data but also in unseen, out-of-sample periods. This matters because a model that overfits past patterns often ends up memorising noise, failing under changing market conditions. Alvi also assessed whether the BN's learned relationships — for example, links between oil prices, volatility indices, and macroeconomic indicators — made economic sense. Stress-testing under different macroeconomic and geopolitical scenarios ensured the models remained robust when the market environment shifted.

Validation Process in GWP 1 and GWP 2

In GWP 1, we adopted Alvi's HMM validation approach by training the model on October 2000–November 2014 data, then applying it to our validation period (December 2014–April 2019). This allowed us to see if the transition probabilities from the training set could still capture regime changes during major events such as the 2014–2016 price collapse recovery and the 2018 market correction. We checked these predicted regimes against movements in macro indicators like VIX, DXY, and GDP growth. In GWP 2, we extended the process to Bayesian Networks, testing whether conditional dependencies — for instance, the increased likelihood of a high-volatility regime when both VIX and DXY rise — persisted beyond the training period. This confirmed that the BN's causal links were not statistical quirks but durable relationships with real-world explanatory value.

From a modeling perspective, validation is the quality control stage that determines whether a model is market-ready. Our results showed that the HMM consistently flagged regime changes under new structural shifts, while the BN preserved economically intuitive relationships in out-of-sample data. Risk metrics like Value-at-Risk and Expected Shortfall remained stable across scenarios, giving us confidence that these models can operate reliably in live conditions — where volatility regimes, macroeconomic drivers, and geopolitical pressures can shift quickly. This alignment of statistical performance with economic logic is what makes them practical tools, capable of supporting real-world trading, risk management, and strategic decisions.

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Step 4 - Replicating results from the Danish A. Alvi paper

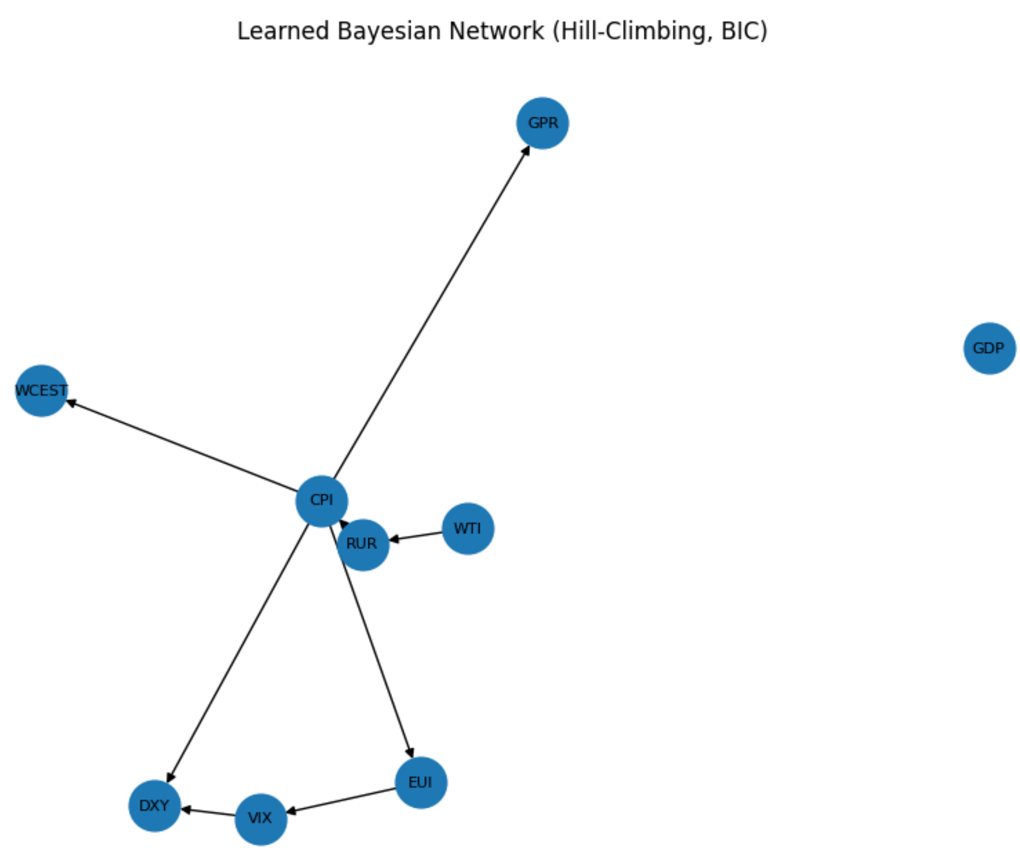
Our Algorithm

Firstly, we removed columns with over 20% of data missing. This allowed us to forward and backward fill remaining missing values and ensure a complete dataset. Our next step was to discretize numeric variables into three quantile bins, and categorical variables were factorized into integer codes. Finally, we removed degenerate variables (only one unique state). The algorithm that was used was Hill-Climbing structure learning with BIC score for discrete CPDs, max three parents per node, and stopping when no improvement was seen.

The Results

Hill-Climbing completed.
Iterations: 9, Final BIC score: -3430.01, Number of nodes: 10, Number of edges: 8

Learned edges (parent -> child):
CPI -> DXY, CPI -> EUI, CPI -> GPR, CPI -> WCEST, EUI -> VIX, RUR -> CPI, VIX -> DXY, WTI -> RUR



In order to check whether we got the same results as the dissertation, we compared learned edges against the network structure presented in Alvi (2018). While the original thesis also used Hill-Climbing and Bayesian Networks, the learned structure here differs in both edge direction and variable connections. This could have happened for a number of reasons, such as a different discretization method, specific search constraints could have been applied in order to 'guide' the network toward a known causal relation, or even due to the use of BDeu.

To conclude, our replication did not fully match the paper's. While the Hill-Climbing procedure successfully produced a plausible Bayesian Network, differences in preprocessing, discretization, and scoring likely account for structural discrepancies. In order to further explore this, we also decided to implement a Hill-Climbing search with BDeu scoring. However, as can be seen from the results, we still have not been able to replicate the paper fully.

Step 5 - Interpretation of results

The Method

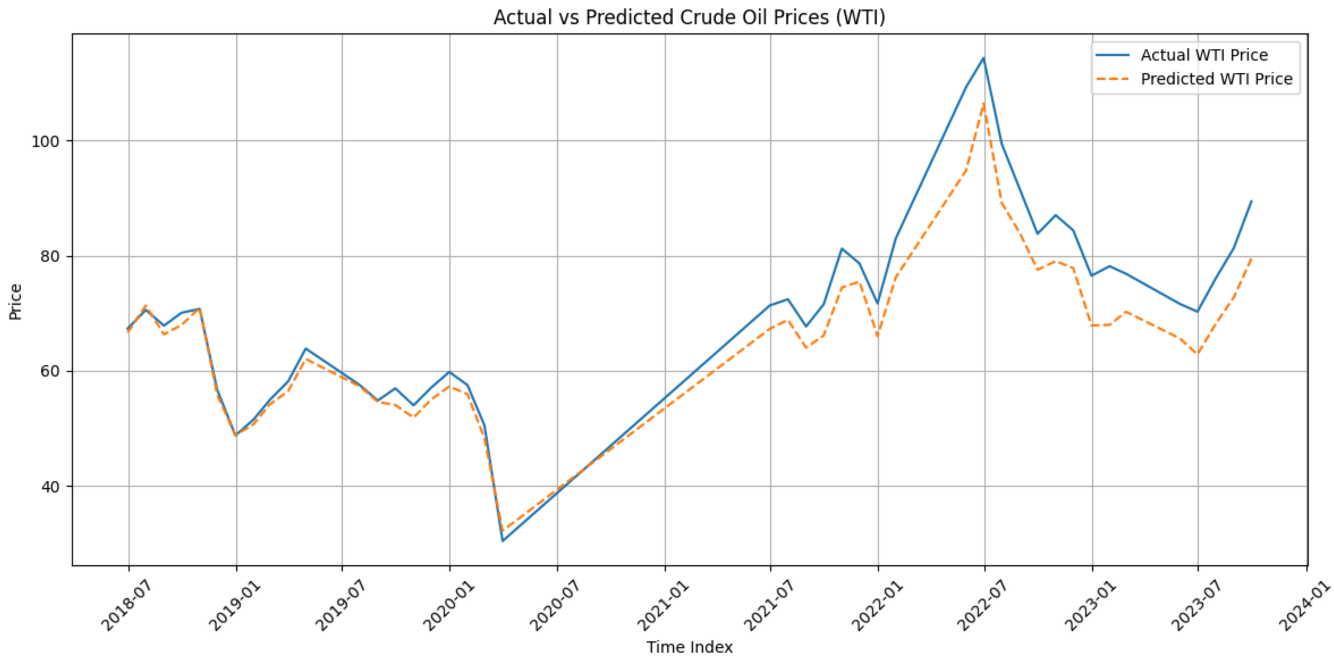
We started by setting WTI as the target variable, and we dropped the rows where it was missing. This allowed us to take all the remaining numeric columns as predictors, filling the remaining missing values with forward- then backward-fill to avoid training errors. We split the data into 80% train and 20% test, so the model learns on "past" and predicts "future" indices. We trained a regression model and evaluated the forecasting accuracy using the MSE and RMSE.

The Results

Our RMSE shows us that our predicted price is off by about \$5.67 from the actual WTI crude oil price. Considering the WTI price range in the plot (roughly \$30–\$110), this is a relatively low error. Our R^2 value shows us that our model explains 87.6% of the variance in the actual WTI prices on the test data, which is strong predictive power for a simple linear regression. Overall, the predictive curve follows the actual price closely, except for a slight underestimation in periods of sharp increase.

This level of accuracy suggests the predictors used have a strong explanatory value for WTI prices.
However, the residual gap during volatile peaks means the model may be less reliable during sudden market shocks.

Adding lagged features (previous month prices, last month macroeconomic indicators) or using a nonlinear model (e.g., Gradient Boosted Trees, Neural Networks) could reduce the underestimation during spikes.



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Step 6 - Assessing contributions

The Alvi paper in Chapter 5, “Conclusion & Future Work,” seems to have some editorial mistakes. In the second paragraph, where he lists the contributions to the existing literature, the point “Secondly” is made twice:

“Secondly, the idea of using time-series data discretised by Hidden Markov Models as inputs to Belief Networks is novel.”
“Secondly, given the structural learning techniques employed, it is almost an autonomous decision-making system which requires absolutely no prior expert knowledge, other than the selection of datasets, which commodity market analysts carry out.” (Alvi 66.)

Another editing mistake is that the first “Secondly” and the “Fourthly” points are simply the same.

“Secondly, the idea of using time-series data discretised by Hidden Markov Models as inputs to Belief Networks is novel.”
“Fourthly, the idea of using time-series data discretised by Hidden Markov Models as inputs to Belief Networks is novel.” (Alvi 66.)

After removing the repeated point about the usage of HMM and assuming that the second “Secondly” point stands on its own and was probably meant as the “Fourthly”, we in fact end up with eight proposed results:

1) Replacing EGARCH-M derived views with Bayesian model-derived views for the Black-Litterman portfolio optimization model

a) Assessing contribution

Alvi’s innovation was to replace the EGARCH-M (Exponential GARCH-in-Mean) model-derived views with the view from the Bayesian probabilistic graphical models when generating the input for the Black-Litterman model.

Alvi argues that one of the main problems of GARCH models - compared to the Bayesian model - is that GARCH models assume that the conditional error variance of spot price follows the ARMA model. Also, the GARCH models are unable to incorporate the macroeconomic, microeconomic, and geopolitical data efficiently, which to a great extent drive the oil prices. (Alvi 2). This new model allows for the generation of investment opinions (views) not only based on volatility and past price moves but also on more fundamental market drivers.

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b) Citations regarding the proposed result

Page 2, Abstract:

"The primary contribution of this dissertation… is robust Bayesian-based models for the oil market to handle challenges in forecasting the spot price of crude oil... Computational Finance and Bayesian Methodologies are proposed…" (Alvi 2)

Page 3, Introduction:

"The first problem is that the existing models for analysing the spot price for crude oil such as GARCH are based on the mere assumption that the error variance of the spot price of crude Oil forecast follows the ARMA model, and therefore these models certainly do not incorporate the macroeconomic, microeconomic and geopolitical events which play an important role in determining the spot price for crude oil. The research proposes using Bayesian-derived views, which can be compared with the GARCH-based views to determine which scheme is better." (Alvi 3)

The above-cited passages state the limitation of EGARCH-M (ARMA error variance, lack of exogenous data) and the intention to shift the source of Black Litterman views to the Bayesian graphical model framework.

c) Reflection on Alvi's work

Do we think that the author accomplished this?

Yes, Alvi has contributed on both the conceptual and implementation levels. The limitations of GARCH models were shown, and the thesis explains the process of building a Bayesian network. Also, this new model is able to source data from a wide range of macro and physical market variables (Chapter 3, methodology).

d) Importance and impact

This contribution is important because it modernizes the way market views are constructed: Instead of relying only on past volatility, the proposed new approach adds up-to-date information about macroeconomic conditions, supply shocks, and geopolitical context.

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2) Using a Hidden Markov Model to discretize time series data as an input to Belief Networks

The paper claims that this integration is novel and implements an HMM-based regime detection for discretization.

a) Assessing contribution

Alvi's dissertation introduces the integration of Hidden Markov Model (HMM) discretized time series into Bayesian (Belief) Networks. The HMMs are used to transform continuous time-series market data (e.g., oil prices, bond prices, macro variables) into regime-labeled discrete states (such as bull, bear, stagnant). These labelled data series are used as input features for learning the structure and conditional probabilities in belief networks.

The novelty lies in the proposed pipeline: Typical Bayesian networks often use either raw or simply binned time series, whereas here, the regime is additionally detected by HMMs. This has the benefit of being able to capture temporal market shifts and nonlinearity. This regime-labelled data is embedded as categorical input, which helps the Bayesian network to track (and utilize) regime-related statistical dependencies.

b) Citations regarding the proposed result

Page 42, Section 3.3.1 (Regime Detection):
"Regime detection is the process by which we identify periods of similar volatility in a time-series model. ... Given that these periods are latent and can only be observed as emissions (returns), the problem ultimately decomposes to the problem that Hidden Markov Models were constructed to solve. ... In our design specification, we have decided to use Hidden Markov Models to identify these regimes in the time-series data." (Alvi 42)

Figure 3.2, Page 43:
The diagram shows a 3-state HMM model that detects bull, bear, and stagnant regimes. (Alvi 43)

Page 53-57, Section 4.2:
Python implementation: *"We would be using a Python library called hmms for implementing the Hidden Markov Models ... The regime detection will comprise four main stages: ... Learning the parameters of the assumed HMM generating the time series ... Finding the most likely sequence of hidden states ... Identifying the latent meaning behind each hidden state ... so that we can discretise our data and use it as an input to our belief network." (*Alvi 53-57)

Page 56, Figure (plot in code):
HMM regimes shown with colors on the WTI oil price lineplot (Alvi 56)

c) Reflection on Alvi's work

The author demonstrates the full programming pipeline in Python and applies it to the dataset. The regime-based sequence is created from the price and macro variables via HMMs. These discrete states are then used to construct the Belief Network dataset. The thesis documents both the conceptual aspects and demonstrates the technical implementation. A limitation of the work is that it does not evaluate how much the regime-aware discretization improves model accuracy.

d) Importance and impact

Introducing economically meaningful states vastly improves the interpretability of the process and the resulting data. Regime detection with the help of HMMs lets us now capture the financial environments better than the previously used binning or raw returns. Ultimately, it enables the design of trading systems that are able to adapt to changing market regimes.

3) Introducing a working trading mechanism that is directly deployable in the commodity markets

This paper presents a workflow and working example of a trading system that uses publicly available macroeconomic, microeconomic, geopolitical, and market data and generates forecasts from it.

a) Assessing contribution

Alvi's paper takes a step beyond just theoretical modeling: It presents a workflow and working solution of an automated trading system for crude oil. His solution uses numerous datasets from macroeconomic, microeconomic, and geopolitical areas. These datasets are preprocessed, discretized (using HMMs), and used as inputs for a Bayesian belief network. Also, this network is trained by automated methods (Hill Climbing, BIC score). This model outputs forecasts and signals that can be used for trading. In summary, his system gives an automated end-to-end solution for data retrieval, cleaning, discretization, network training, and forecasting, needing minimal expert input. Except for the initial dataset selection, the network and other parameters are learned from data.

b) Citations regarding the proposed result

Page 2, Abstract & Contributions: *"...robust Bayesian-based models for the oil market to handle challenges in forecasting the spot price of crude oil in a real-world trading environment for portfolio managers…" (Alvi 2)*

Page 48, Section 4, Implementation:
"The document contains the implementation of the oil trading system using graphical models…" (Alvi 48)

Page 65, Results and Comparison:
"As we can see, the algorithm has successfully managed to hedge the risk in the early quarter of 2015, but did end up shorting in a bull run in late 2017/early 2018." (Alvi 65)

Page 66, Conclusions:
"...it provides an automated trading mechanism which learns and improves its trading decisions as time passes, ultimately resulting in a higher alpha for a commodities trader." (Alvi 66)

c) Reflection on Alvi's work

As per our evaluation, the author has accomplished this goal because every major aspect is implemented with code, documentation, and results. The code is modular and divided into sections for data retrieval, cleaning, discretization, model training, validation/testing, and forecasting. The implementation uses well-known open-source libraries (pgmpy for PGMs, hmms for HMM regime detection). The system can also run autonomously after the initial dataset selection. Further improvement of the solution could have been to include empirical performance comparisons.

d) Importance and impact

This work includes both theory and practice by providing a concrete, working solution for commodity trading. It also demonstrates that automation and objectivity can be introduced by using network learning, which reduces the need for expert input. This reduces the chance of bias and human oversight, resulting in a more consistent system. Due to the modular architecture, this solution can be adapted to other commodities as well, making it highly impactful in the field of quantitative finance. The idea of integrating macroeconomic driving factors and using regime-aware modeling helps with the interpretability of the machines compared to the usual black-box machines.

4) Introduction of a Bayesian structure learning method to alleviate the need for human experts

The huge amount of datasets poses a considerable challenge for quant firms, because they usually need expert knowledge of the datasets to find relations between them. This manual approach is not just time-consuming, but also error-prone. To solve this issue, the paper proposes a system to learn the Bayesian Network from the data.

a) Assessing contribution

Alvi's work solves the challenge of limited availability and high cost of expert input when analyzing commodity market data. The large volume of the data and complexity of macroeconomic, microeconomic, and geopolitical datasets pose a huge challenge (and cost) when the mapping of the relationships is carried out manually. This manual effort is not just costly but also prone to bias and error. The automated Bayesian structure learning algorithms learn the dependency structure directly from data. This automated approach helps tremendously with reproducible network discovery.

b) Citations regarding the proposed result

Page 3, Motivation & Problem Statement:
"The research proposes a scheme which would allow the system to learn the Bayesian Network in an attempt to causally relate all datasets without the presence of an expert." (Alvi 3)

Page 17, Section 2.1.1.1.2.1:
"In the next section, we will be exploring the process of learning a belief network using score-based learning and will be describing different optimisation techniques such as the Hill Climbing algorithm for constructing the belief networks." (Alvi 17)

Pages 17-21, Section 2.1.1.1.2.2:
Figure 2.4, p.19: This figure visualizes the iterative nature of hill climbing (Alvi 19)

Page 44, Section 3.4, Model Design:
"We therefore propose using structure learning algorithms for constructing a causal network of the oil markets... A popular score-based structure learning algorithm is Hill Climbing…" (Alvi 44)

c) Reflection on Alvi's work

Alvi realizes both the conceptual framework and implementation. His thesis explains the possible downsides of the manual/expert-based approach and shows that there is a huge need for automated structure learning. The technical chapters show the step-by-step implementation using real market datasets and working code examples for structure discovery.

d) Importance and impact

This part of Alvi's work is also very impactful and thus important. Removing the need for manual labor done by highly paid experts increases the quality and reproducibility while significantly lowering the cost. As we incorporate more and more datasets in our model, these benefits become even more significant. It is also worth noting that his proposed solution can be generalized and used for other commodities and markets as well.

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5) Design insight via using Python graphical modeling modules

The abstraction provided by these Python libraries helped clarify the design process.

a) Assessing contribution

Alvi's dissertation uses Python-based graphical modeling libraries like pgmpy for Bayesian/Belief Networks and hmms for Hidden Markov Models. He claims that these libraries provide significant benefits in terms of design clarity and workflow efficiency for the given tasks.
The libraries have a well-developed abstraction layer that lets us transform regime detection and network learning into code modules. This helps with rapid prototyping.

b) Citations regarding the proposed result

Page 28-29, Section 2.2:
"We would precisely be using pgmpy to construct the belief networks for the oil trading model. There are a number of methods for learning the belief networks by learning the data, and we will be exploring such techniques offered by pgmpy. We would also be exploring how to use expert data from the Energy Information Administration and the FRED, and learn our model using that data." (Alvi 28)

Page 48, Chapter 4 (Implementation):
"The document contains the implementation of the oil trading system using graphical models…" (Alvi 48)

Sections 4.1 to 4.3 (pp.48-65) show end-to-end implementation using these modules.
Figure 3.1, Page 35: The Crude Oil Trading Model design workflow

c) Reflection on Alvi's work

The dissertation demonstrates the benefits of using Python libraries. The workflow is documented as a sequence of modular tasks (retrieval, cleaning, discretization, modeling, validation) where each step is implemented using high-level functions from pgmpy and hmms libraries. This makes the workflow easy to follow and adapt if needed. The code is also supported by markdown and comments.

d) Importance and impact

This result from Alvi's work is significant because it accelerates rapid prototyping and lowers technical barriers. It also helps with reproducibility and transparency, because thanks to the well-known open-source libraries, the results can be replicated (or audited). These benefits allow for easier collaboration.

6) Systematic, event-driven forecasting that incorporates geopolitical changes and macroeconomic data and is capable of higher returns

According to Alvi (67), his proposed method is capable of higher returns than HFT or fixed-income strategies.

a) Assessing contribution

Alvi proposes a systematic and event-driven method for forecasting crude oil prices. In his work, he incorporates macroeconomic and geopolitical indicators along with microeconomic and market data. The probabilistic graphical model (PGM) can quickly respond to policy shifts, macroeconomic changes, and critical geopolitical changes.
While HFT and fixed income strategies exploit minute-by-minute (or faster) trends, they usually fail to make use of large, regime-changing shifts. The PGM model can react to these changes, and thus it promises higher returns.

b) Citations regarding the proposed result

Page 2, Abstract and Contributions:
"...robust Bayesian-based models for the oil market to handle challenges in forecasting the spot price of crude oil in the real world trading environment…" (Alvi 2)

Page 65, Results: Comparison plot (part of the code with no figure number) and the commentary section illustrate a case where the model has beaten the performance of the EIA forecast.

c) Reflection on Alvi's work

This goal was mainly achieved, but not completely. The model is proven to be able to support systematic, event-driven decision-making. The structure-learning algorithm used is also capable of capturing the impact of macroeconomic and geopolitical changes. The results (Alvi 65) display trades that successfully hedge or outperform the proposed baseline of the EIA forecast. However, the claim of higher returns than HFT or fixed-income funds is not thoroughly documented. Moreover, the examination of this claim is not statistically rigorous.

d) Importance and impact

What is the importance of this achievement? This result from Alvi's work improves the explanatory power considerably because it explains why price changes occur due to given macroeconomic and geopolitical drivers. If one is anticipating a geopolitical change, then this model can even be used to run different outcome scenarios for the geopolitical change. There is also a potential for higher returns and more effective risk management due to the broader input data.

7) Allowing for the construction of better energy-market models with policy analysis in focus

a) Assessing contribution

The paper shows the value of probabilistic graphical models (PGMs) for constructing energy-market models that are appropriate and usable for policy analysis. By discovering the causal relationships among macroeconomic, microeconomic, and geopolitical variables, PGMs provide a framework to understand oil market dynamics better. With this knowledge, policymakers are able to identify key driving factors in energy markets. This opens up a world of possibilities to simulate the impact of a new policy or assess the risks of a geopolitical event.

b) Citations regarding the proposed result

Page 3, Abstract (Contributions to Science):
"...allows us to understand how the oil markets work and therefore assists economists and energy policy makers to determine which factors to control and what policies to draft in an attempt to prevent the economy from going into a recession and to control climate change by reducing dependency on fossil fuels." (Alvi 3)

Page 67, Conclusions & Future Work: *"...allows us to construct better models of energy markets, hence allowing energy policy makers to understand the underlying structure of the oil markets and use it to respond to different economic events by drafting more effective and sound policy."* (Alvi 67)

c) Reflection on Alvi's work

In our assessment, the Author accomplished this goal. The thesis shows how PGMs are able to map complex data into causal graphs, which then can be used by policymakers to understand the effects of policy changes like OPEC decisions, import tariff changes, sanctions, etc., inventory changes, or macroeconomic shifts.

d) Importance and impact

In complex structures and systems, the biggest challenge is to track changes caused by some intervention. Usually, the complexity of such a system does not allow for a clear assessment of the effect of the introduced change. Therefore, having structural clarity is key for policymakers. Moreover, this is where PGMs shine: they help visualize direct and indirect effects.

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8) Incorporating a number of research studies from multiple disciplines for improved alpha detection for commodity traders

A key contribution, according to the paper, is that it allows for better alpha in the commodities market.

a) Assessing contribution

Alvi's dissertation states that it brings value not just in technical modeling aspects or in a single academic domain. It also integrates research and techniques from other disciplines like economics, computer science, and machine learning. His model utilizes macroeconomic theory and government open data, Bayesian network design (computational statistics), regime detection (econometrics), and modularly coded workflows (computer science).

b) Citations regarding the proposed result

Page 3-4, Objective of Research
"By using Probabilistic Graphical Models and applying various Machine Learning and Statistical techniques, the accuracy of existing quantitative models forecasting the spot price of oil can be improved by taking into account many different macroeconomic and geopolitical variables in consideration when making a prediction system." (Alvi 3)

Page 67, Conclusion:
"...this research is amalgamating a multitude of existing research and experimentation in multiple disciplines and applying it in the commodity markets…" (Alvi 67)

c) Reflection on Alvi's work

In our opinion, Alvi rightfully claims to have achieved an interdisciplinary fusion of methods and tools. His workflow combines regime detection, policy simulation, Bayesian structure learning, out-of-sample validation, and direct trading applications. He also uses microeconomic, macroeconomic, and geopolitical data.

d) Importance and impact

His approach shows innovation by integrating methodologies and theories across multiple domains. This can reveal new alpha sources, which is a very considerable achievement in this competitive quantitative finance space.

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Step 7 - Non-technical summary

Discussion about what this study does better, quicker, or more cost-effectively than the existing prediction models.

This study has introduced a new, data-driven approach to forecast crude oil prices using probabilistic graphical models (PGMs). Compared to the traditional prediction model, this solution stands out in the following ways:

Integration of real-world drivers

Most older models solely focus on historical price and volatility patterns, which is like looking at temperature values without context. In contrast, this new proposed model is aware of the fundamental driving forces from macroeconomics, microeconomics, and geopolitical fields, which is comparable to satellite images where the principal driving forces of the temperature are better seen. With this, one is better able to understand the causal structure of the oil market.

Using adaptive regime detection

The proposed model does not rely on the assumption that the markets are always the same. By incorporating HMM-based regime detection, this model is able to better adapt to market-regime changes. This is important because the investors and thus the market behave significantly differently in bull markets than in bear markets. Traders and policy-makers can now adapt their strategies to be tailored to the existing market regime.

Minimizing the need for expert input - cost effectiveness and scalability

Traditional oil forecasting has relied heavily on expert knowledge and intuition in the past. This introduced at least three major issues:

- Experts are costly, and as the dataset increases, this becomes exponentially true
- Human intuition is not unbiased
- Human intuition is not reproducible due to a lack of complete objectivity

These issues have been alleviated by the use of network learning, where the structure is directly learned from data. This method is faster, cheaper, and less error-prone, which means, in summary, that it is easier to scale.

Automated and deployable workflow

The proposed modeling pipeline - starting from data retrieval to regime detection, network construction, and forecasting - is implemented in Python code with open-source libraries. This allows for rapid prototyping and is more accessible to researchers due to lower technical and financial barriers (no proprietary code is used).

Event-driven prediction with higher possible returns

Compared to legacy models, which lag behind market events by only watching price and volatility data, this framework is able to react to geopolitical and economic changes. The backtest showed that the model outperformed the EIA forecasts even during market disruptions.

Enhanced policy analysis and transparency

In the past, the forecasting models were either black-boxes (Neural Networks are usually obscure by nature) or simpler models that lacked modern ML methods. The proposed method in this paper utilizes ML techniques in a way that retains transparency. In fact, with the network structure learning, it actually discovers relationships between the underlying factors and the oil price. This can help policymakers better track the effect of the different changes in policy.

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